We get, $\frac{dy}{du} = 3u^2 = 3\sin^2 x$

$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = 3\sin^2 x \cdot \cos x$

(vi) $y = \sqrt{x^2 + x + 1}$

Let $u = x^2 + x + 1$ then $y = \sqrt{u}$

Differentiating $u = x^2 + x + 1$ with respect to x

We get, $\frac{du}{dx} = 2x + 1$

Differentiating $y = \sqrt{u}$ with respect to u.

We get, $\frac{dy}{du} = \frac{1}{2\sqrt{u}}$

$= \frac{1}{2\sqrt{x^2 + x + 1}}$

$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \frac{1}{2\sqrt{x^2 + x + 1}} \cdot (2x + 1)$

$= \frac{2x + 1}{2\sqrt{x^2 + x + 1}}$

DO YOU KNOW?

★ $\frac{d}{dx}(\sin(\log x))$

$= \frac{d}{d(\log x)}(\sin(\log x)) \frac{d}{dx}(\log x)$

★ $\frac{d}{dx}(x^2 + 3x + 1)^5$

$= \frac{d}{d(x^2 + 3x + 1)}(x^2 + 3x + 1)^5 \frac{d}{dx}(x^2 + 3x + 1)$

★ $\frac{d}{dx}(e^{(x^3+3)})$

$= \frac{d}{d(x^3 + 3)}(e^{(x^3+3)}) \frac{d}{dx}(x^3 + 3)$

Example 5.34

If $f(x) = x^n$ and $f'(1) = 5$, then find the value of n.

Solution

$f(x) = x^n$

$\therefore f'(x) = nx^{n-1}$

$f'(1) = n(1)^{n-1}$

$f'(1) = n$

$f'(1) = 5$ (given)

$\Longrightarrow n = 5$

Example 5.35

If $y = \frac{1}{u^2}$ and $u = x^2 - 9$, then find $\frac{dy}{dx}$.

Solution

$y = \frac{1}{u^2} = u^{-2}$

$\frac{dy}{du} = -\frac{2}{u^3}$

$\frac{dy}{du} = -\frac{2}{(x^2 - 9)^3}$ $(\because u = x^2 - 9)$

$u = x^2 - 9$

$\frac{du}{dx} = 2x$

Now, $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

$= -\frac{2}{(x^2 - 9)^3} \cdot 2x$

$= -\frac{4x}{(x^2 - 9)^3}$

Exercise 5.5

1. Differentiate the following with respect to x.

(i) $3x^4 - 2x^3 + x + 8$

(ii) $\frac{5}{x^4} - \frac{2}{x^3} + \frac{5}{x}$

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